

Develop Generative AI Applications: Get Started

Today's Skill Points

Module 1

Foundations of Generative AI and Prompt Engineering

Welcome to the Course

- Course Introduction
- Mid and End-of-Course Certificate Overview
- Course Overview
- Mid and End-of-Course Completion

Optional Generative AI Essentials

- About this Optional Lesson
- Introduction to Generative AI
- What are Generative AI Models?
- What is a Natural Language Processing?
- Reading Comprehension Guide to Generative AI
- Practical Quick Generative AI Course

Working with Prompt Engineering and Prompt Templates

- Introduction to Prompt Learning
- Introduction to LangChain
- Advanced Methods of Prompt Engineering
- LangChain LCEL Chaining Method
- Optional Reading: What is Prompt Engineering and Why You Need?
- Choosing the Right Prompt Strategy: A Practical Checklist
- Lab: Master Prompt Engineering and LangChain Prompt Templates
- Practical Quick Working with Prompt Engineering and Prompt Templates

Module Summary and Evaluation

- Summary and Insights: Foundations of Generative AI and Prompt Engineering
- Check Your Progress: Foundations of Generative AI and LangChain
- Graded Quick Foundations of Generative AI and Prompt Engineering

Module 2

Introduction to LangChain in GenAI Applications

Module 3

Build a Generative AI Application with LangChain

Course Overview

1 of 3 Learning Objectives (Foundations of Generative AI)

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Hi, and welcome to the **Develop Generative AI Applications: Get Started** course.

This course is designed for aspiring AI developers, machine learning engineers, data scientists, and AI researchers. It equips you with the knowledge and skills to develop and integrate generative AI solutions. You'll explore core prompt engineering techniques, such as in-context learning, and learn to create and manage prompt templates using frameworks like LangChain.

As you progress, you'll master key LangChain concepts, including chains, tools, and agents, to build smarter AI applications. Through hands-on labs, you'll experiment with AI model configurations and develop functional AI-powered applications. By the end, you'll have the expertise to design and implement generative AI solutions effectively.

This course is part of the [GenAI and Prompt Engineering Professional Certificate](#), designed to provide you with the practical skills and knowledge to build advanced AI applications that leverage retrieval-augmented generation (RAG), multimodal AI, and agentic AI systems.

Prerequisites

Python programming skills and experience are required, as the course emphasizes coding and application development. Additionally, fundamental knowledge of AI and web development is highly recommended. A strong foundation in generative AI is recommended but not required.

Objectives

After completing this course, you will be able to:

- Examine foundational concepts of generative AI and the LangChain framework, focusing on how prompt engineering and in-context learning enhance AI interactions.
- Apply prompt templates, chains, and agents to create flexible and context-aware AI applications using LangChain's modular approach.
- Develop a generative AI web application with Flask, integrating advanced features such as JSON output parsing for structured AI responses.
- Evaluate and compare different language models to select the most suitable option for specific use cases, ensuring optimal performance and reliability.

Course Outline

This course consists of three modules.

Module 1: Foundations of Generative AI and Prompt Engineering

- Lesson 0: Welcome
- Lesson 1 (Optional): Introduction to Generative AI
- Lesson 2: Working with Prompt Engineering and Prompt Templates
- Lesson 3: Module Summary and Evaluation

Key Topics:

- Generative AI Models
- NLP (Natural Language Processing)
- Prompt Engineering and In-Context Learning
- Advanced Methods of Prompt Engineering
- LangChain LCEL Chaining Method
- LangChain Prompt Templates

Module 2: Introduction to LangChain in GenAI Applications

- Lesson 1: LangChain Core Components and Advanced Features
- Lesson 2: Module Summary and Evaluation

Key Topics:

- LangChain Core Concepts
- LangChain Advanced Features: Chains and Agents
- Build Smarter AI Apps: Empower LLMs with LangChain

Module 3: Build a Generative AI Application with LangChain

- Lesson 1: Application Development Workflow with Generative AI
- Lesson 2: Module Summary and Evaluation
- Lesson 3: Course Wrap-Up

Key Topics:

- Choose the Right AI Model for Your Use Case
- From Ideas to AI: Building Applications with Generative AI
- Python with Flask for Large-Scale Projects
- Hands-on with GenAI: Choosing the Right Model for Your Application

Tools/Software

In this course, you will explore and utilize a variety of tools and platforms, including:

- LangChain to design and implement structured AI workflows, integrate LLMs with external data sources, and optimize prompt engineering techniques.
- Flask, HTML5 and CSS3 to develop AI-powered web applications with interactive features.
- LangChain's `JsonOutputParser` to generate structured JSON outputs from LLM responses for enhanced data usability.
- LLMs (Llama 3, Gemin, Mistral) to experiment with different AI models, compare their performance, and select the best fit for specific applications.
- Python for coding AI applications, integrating APIs, and implementing LangChain's functionalities effectively.

Tips for Success

- Stay organized and track key program milestones, risks, and stakeholder communications.
- Consistently practice using the generative AI tools and techniques covered in the course.

Congratulations on taking this step to elevate your career with generative AI! Enjoy your learning journey.

Mark as completed

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